

Excitation-Coherent Telemetry Analysis for Predictive OBD-II Connector Wear Detection

Real telemetry from 5 public datasets · 9 vehicles in detail + 381-vehicle fleet · vehicle-disjoint validation

Problem

Telematics / insurance dongles stay plugged into the J1962 port for years. Vibration frets the tin-plated contacts: milliohms → intermittent ohms. The car never notices. The dongle sees lost frames, timeouts and brown-outs, and gets replaced while the worn connector stays in service.

Method: ECTA

Uses only what the tool already records.

- 1 Loss features (self-referenced)
- 2 Loss-structure features (selectivity, lateness, load)
- 3 Excitation coherence: do losses rise with the car's own rpm and speed?
Poisson GLM $y \sim \exp(\alpha + \beta \cdot \log v)$
Physics model + first-principles observability law.

Results

Unseen vehicles (full-bus, 6-fold LOVO):

AUROC 0.880; moderate 0.891, severe 0.995; deep baselines ≤ 0.65

Moderate wear confirmed in ~2 min

Observability law predicts

detectability: Spearman $\rho = 0.93$

Holds on a different fault model (0.856)

Fleet: 88 % timely warnings, 1.5 % false alarms

No added hardware · open, reproducible code · limits reported (incipient wear, low-rate tools, long-horizon RUL)